

Multistate Torsion in *Zitterbewegung*

Wojciech J. Jankowski^{1,*}

¹*TCM Group, Cavendish Laboratory, Department of Physics,
J J Thomson Avenue, Cambridge CB3 0HE, United Kingdom*

(Dated: September 30, 2026)

We derive nonlinear *Zitterbewegung* velocity of electrons augmented by band torsion in crystals. The equation of motion with multistate quantum geometric tensors and torsion culminates in transient nonlinear photocurrent responses distinct from the recently measured resonant steady-state photovoltaic effects induced by circularly polarized light. Tomography protocols for measuring the multistate quantum geometric torsion tensors are proposed.

Introduction.— Quantum geometry in condensed matter physics provides a versatile paradigm for studying multiband effects in solids [1–5]. Applications of the quantum geometric frameworks range from superconductivity [6–8], magnetism [9–12], or excitonics [13–17], to linear and nonlinear transport and optics [18–22]. In spirit, these frameworks can be systematically reduced to identifying gauge-invariant quantum tensors defined by inner products of quantum state projectors and their derivatives [22, 23] within the perturbative structure of quantum response functions. This approach, underpinned by identification of quantum geometric bounds [6–8, 14, 24–31], has proven particularly fruitful for studying physical consequences of nontrivial band topologies in solids, through their quantum geometric fingerprints.

At heart of the quantum geometric frameworks is the study of the quantum geometric tensors (QGTs) [32–34], also known as Hermitian metrics of quantum states [5, 22]. As complex quantities, QGTs consist of a real part, the quantum metric [32], and imaginary part, the Berry curvature equal to a geometric Berry phase over an infinitesimal loop in parameter space [35, 36]. The parameter space of choice in condensed matter systems is provided by crystal momenta $\mathbf{k}$ of Bloch states over the Brillouin zone (BZ). As gauge-invariant objects, the QGTs resolved over the momentum space have been measured in crystals only recently [37, 38].

The nontrivial QGTs definitionally require a single quantum state, and a Hilbert space with at least two states [32]. However, it is further possible to define gauge-invariant two-state [22, 23] and multistate QGTs beyond Berry curvature and quantum metric, such as band torsion [21, 22, 39, 40], which arise in nonlinear optics [21, 22, 41–43] and transport phenomena [39, 40] involving multiple energy levels. The interest in multistate quantum geometries has been particularly reinvigorated [39, 42, 44] with the discoveries of exotic nonlinear photovoltaic and viscoelastic effects involving higher-rank QGTs and band torsion [41, 45, 46]. Notably, photovoltaic circular shift photoconductivities associated with band torsion have been experimentally measured in MnPSe₃ antiferromagnetic crystals very recently [47].

On a seemingly different end, QGTs have been most recently revisited from the perspective of classical Hamiltonian dynamics in presence of effective magnetic fields [48], subject to semiclassical quantizations of angular momentum. Yet, this opens up a question about the interpretations and the role of multistate QGTs and band torsion in semiclassical dynamics [22, 23, 26, 41, 49].

In this work, we address this question from an angle of the semiclassical equations of motion for electron velocity in clean crystals. In doing so, we surprisingly find uncharted transient effects driven by band torsion. First, we demonstrate that the multistate QGTs and band torsion arise in the nonlinear *Zitterbewegung* velocity associated with nonequilibrium jittery trembling motion of electrons under external electric fields. Second, we show that these multimode *Zitterbewegung* velocities of electrons culminate in transient nonlinear photocurrents due to the multistate torsion tensor. These photocurrents may be measured in an instantaneous nonlinear response to circularly polarized optical fields. Finally, we argue that the multimode *Zitterbewegung* enables a dynamical tomography of band torsion, allowing one to access this quantum geometric quantity directly, in contrast with the resonant response measurements. These findings apply to transient bulk responses of ultraclean crystalline materials and are further adaptable for ultracold atom quantum simulators.

Multiband Zitterbewegung.— Before deriving the nonlinear multiband *Zitterbewegung* central to this work, we begin with the field-induced first-order *Zitterbewegung* contributed by an interband coherence associated with the two-state quantum geometry. At first order in electric fields E_j , the equation for an electron velocity $v_i^m(t)$ in a nondegenerate band with index m in a clean crystal reads [50],

$$v_i^m(t) = \frac{1}{\hbar} \partial_i \varepsilon_m + \frac{e E_j}{\hbar} \Omega_{ij}^m + \xi_i^m(t). \quad (1)$$

The momentum-space derivative, $\partial_i \equiv \partial_{k_i}$, with $i, j = x, y, z$, acts on the band energy ε_m in a dispersive drift velocity term, and Berry curvature Ω_{ij}^m contributes a linear Hall drift of an electron. The last term represents the static-field-induced first-order *Zitterbewegung*

* wjj25@cam.ac.uk

velocity [50],

$$\xi_i^{m,(1)}(t) = \frac{ieE_j}{\hbar} \sum_{n \neq m} Q_{ij}^{nm} e^{-i\omega_{mn}t} + \text{h.c.}, \quad (2)$$

with frequency $\omega_{mn} = (\varepsilon_m - \varepsilon_n)/\hbar$, two-state QGT, $Q_{ij}^{nm} = A_i^{nm} A_j^{mn}$, and non-Abelian Berry connection, $A_i^{nm} = i\langle u_{n\mathbf{k}} | \partial_i u_{m\mathbf{k}} \rangle$, derived from Bloch states $|u_{m\mathbf{k}}\rangle$. Upon promoting the static electric field E_j to an optical field $E_j(t) = \text{Re}[E_j(\omega)e^{i\omega t}]$, a dynamical field-induced first-order *Zitterbewegung* velocity arises, as we show in the Supplemental Material (SM) [51]. The dynamically-induced *Zitterbewegung* provides a microscopic physical interpretation of the charge dipole correlations in the framework of time-dependent quantum geometry [52].

Centrally to this work, at second order in the optical electric fields $E_j(\omega)$, a nonlinear multimode *Zitterbewegung* velocity further arises,

$$\xi_i^{m,(2)}(t) = \frac{e^2 E_j(\omega) E_k(\eta\omega)}{\hbar^2} \sum_{a,b=n,p,m} \mathcal{A}_{ijk}^{ab}(\omega, \eta\omega) e^{-i\omega_{ab}t}, \quad (3)$$

with $\eta = \pm$ corresponding to distinct frequency combination channels. The frequency-dependent factors $\mathcal{A}_{ijk}^{ab}(\omega, \eta\omega)$ encode the two-state and three-state QGTs, $Q_{ijk}^{nmp} = A_i^{nm} A_j^{mp} A_k^{pn}$, as derived in the SM [51]. Interestingly, a nonlinear multimode *Zitterbewegung* velocity in band m includes modes with frequencies ω_{pn} such that $p, n \neq m$ correspond to a distinct pair of virtual or unoccupied levels. As a key result, we demonstrate that the multistate band torsion, $T_{ijk}^{nmp} = i(Q_{ijk}^{nmp} - Q_{ikj}^{nmp})$, arises in circular *Zitterbewegung* contributed by modes with frequencies ω_{pn} independent of the occupied band m . As a consequence, the band torsion is directly manifested in the high-frequency modes, with $\omega_{pn} > \omega_{mn}, \omega_{mp}$, of transient photocurrents.

Nonlinear Zitterbewegung currents.— Having derived the second-order *Zitterbewegung* velocities, we focus on the associated transient photocurrents, which can be induced by an instantaneous application of optical fields.

The bulk electric current j_i can be computed from density matrix $\hat{\rho}$ and velocity operator $\hat{v}_i$,

$$j_i(t) = -e \text{Tr}[\hat{\rho} \hat{v}_i] = -e \sum_{n,m,\mathbf{k}} \rho_{nm} v_i^{mn}. \quad (4)$$

For $n \neq m$, $v_i^{nm} = i\omega_{nm} A_i^{nm}$, and the single-particle density matrix interband coherences ρ_{nm} can be perturbatively expanded as $\rho_{nm} = \rho_{nm}^{(0)} + \rho_{nm}^{(1)} + \rho_{nm}^{(2)} + \dots$, while satisfying quantum Liouville equation [50, 53]. Centrally to this work, we consider the interband coherences up to the second order, $\rho_{nm}^{(2)}$, in the optical fields. The multistate band torsion arises in the three-state contributions to $\rho_{nm}^{(2)}$, which translate into the nonlinear *Zitterbewegung* velocity, $\xi_i^{m,(2)}(t)$, as demonstrated in the SM [51].

The second-order field-induced *Zitterbewegung* currents which arise in transient nonlinear responses read,

$j_i^{(2)}(t) = -e \sum_m f_m \xi_i^{m,(2)}(t)$. Centrally to this work, these currents include multistate band torsion T_{ijk}^{nmp} contributions in instantaneous responses to electric fields with magnitude E_0 ,

$$j_i^{(2)}(t) \Big|_{\omega_{pn}} = -\frac{e^3 E_0^2}{\hbar^2} \sum_{m,n,p,\mathbf{k}} f_m w_{nmp}^{+-}(\omega) \text{Im} [T_{ijk}^{nmp} e^{-i\omega_{pn}t}]. \quad (5)$$

with the off-resonant frequency-dependent factors $w_{nmp}^{+-}(\omega) = \frac{\omega(\omega_{pm}^2 - \omega_{mn}^2)}{(\omega_{mn}^2 - \omega^2)(\omega_{pm}^2 - \omega^2)}$. In the above, we consider the counter-propagating channel, $\eta = -$, and a response to the left circularly polarized light with an electric field vector, $\mathbf{E}(t) = \Theta(t)E_0[\hat{x} \cos(\omega t) + \hat{y} \sin(\omega t)]$, $\Theta(t)$ the unit Heaviside step function, and $\hat{x}, \hat{y}$ the unit vectors in the x and y directions. For the derivation of the corresponding *Zitterbewegung*, see SM [51]. Notably, under the linear polarization, $\mathbf{E}(t) = \Theta(t)E_0[\hat{x} \cos(\omega t)]$,

$$j_i^{(2)}(t) \Big|_{\omega_{pn}} = -\frac{e^3 E_0^2}{\hbar^2} \sum_{m,n,p,\mathbf{k}} f_m w_{nmp}^{\hat{x}\hat{x}}(\omega) \text{Im} [Q_{ixx}^{nmp} e^{-i\omega_{pn}t}], \quad (6)$$

with the distinct off-resonant frequency factors, $w_{nmp}^{\hat{x}\hat{x}}(\omega) = \sum_{\eta=\pm} \frac{\omega_{pn} - \eta\omega}{(\omega_{pm} - \eta\omega)(\omega_{mn} - \eta\omega)}$. Therefore, the nonlinear *Zitterbewegung* currents within a second-order transient response to linearly polarized optical electric fields allows to probe the three-state QGTs Q_{iaa}^{nmp} with $a = x, y$. Thus, we have shown that controlling the polarization of optical fields allows to determine distinct multistate geometric tensors.

Discussion and conclusion.— We further discuss the applicability and significance of derived results.

First, we establish the second-order circular *Zitterbewegung* velocities and the associated transient electric currents as the direct smoking-gun probes of band torsion. It should be stressed that, contrary to the derived circular *Zitterbewegung*, the band torsion, $T_{ijk}^{mn} = \sum_{p \neq n,m} T_{ijk}^{mnp}$, does not arise in the individual resonant ($\omega = \omega_{mn}$) circular shift photoconductivity tensor components *per se*. In fact, the band torsion contributes only the totally antisymmetric part of the circular shift photoconductivity tensor, $\mathcal{H}_{xyz}^{nm} \equiv T_{[xyz]}^{nm}$ [41, 49, 54], where [...] denotes an antisymmetrization in all indices. Therefore, the band torsion cannot be directly reconstructed from the single-component measurements of resonant circular shift currents, while the momentum space average of its totally antisymmetric part can only be extracted after measuring multiple components of the shift photoconductivity tensor ($\sigma_{xyz}^{\text{shift,C}}, \sigma_{yzx}^{\text{shift,C}}, \sigma_{zxy}^{\text{shift,C}}$) [41]. It should be stressed that, on the contrary, the retrieved *Zitterbewegung* currents under circular drive allow to resolve individual torsion tensor components directly. Therefore, this work establishes the second-order *Zitterbewegung* currents as the first-of-a-kind quantity allowing to directly access the quantum geometric multistate band torsion components, without requiring resonant drive.

Second, we compare the retrieved nonlinear multiband *Zitterbewegung* with the other known quantum trembling motions of electrons in crystals. In the absence of electric fields, the *Zitterbewegung* amplitudes Z_{nm} determined by the interband coherences and captured by non-Abelian Berry connections A_i^{nm} [55, 56], arise in the off-diagonal elements of position operators expressed in the band basis. Hence, the *Zitterbewegung* amplitudes Z_{nm} yield vanishing expectation values upon projecting the electron onto a pure band m . At first order in electric fields, the *Zitterbewegung* velocity magnitude $|\xi_i^{m,(1)}|$ is proportional to the two-state QGTs, $Q_{ij}^{nm} = A_i^{nm} A_j^{nm} \propto Z_{nm}^2$, consistently with Eq. (2) [50, 56]. The interpretation of QGT as field-induced *Zitterbewegung* provides a phenomenological picture for the recent proposal of measuring quantum metric from an instantaneous step response [50, 57]. At second order in electric fields, centrally to this work, $Q_{ijk}^{nmp} = A_i^{nm} A_j^{mp} A_k^{pn} \propto Z_{nm} Z_{mp} Z_{pn}$, which provides a cubic in *Zitterbewegung* amplitudes, interpretation to the second-order field-induced *Zitterbewegung* velocity magnitude $|\xi_i^{m,(2)}|$ in Eq. (3), with $A_{ijk}^{np}(\omega, \eta\omega) \propto Q_{ijk}^{nmp}$. On the other hand, the multistate torsion $T_{ijk}^{nmp} \propto Q_{ijk}^{nmp} - Q_{ikj}^{nmp}$ is represented by the antisymmetrization of *Zitterbewegung* amplitudes picked by the circularly polarized light coupling. At third order in electric fields, we expect a four-state QGT, $Q_{ijkl}^{nmpq} = A_i^{nm} A_j^{mp} A_k^{pq} A_l^{qn} \propto Z_{nm} Z_{mp} Z_{pq} Z_{qn}$, which is quartic in *Zitterbewegung* amplitudes, to determine the field-induced third-order *Zitterbewegung* velocity $|\xi_i^{m,(3)}|$. Phenomenologically, $\xi_i^{m,(3)}$ underpins the quantum geometric enhancement of hyperpolarizability $\chi^{(3)}$ [29]. Hence, the discussed findings establish a hierarchy of *Zitterbewegung* amplitude scalings in nonlinear quantum geometric responses.

Third, we discuss the measurability of the nonlinear *Zitterbewegung* effects driven by the multistate torsion. Although generally underexplored in real materials, the multimode *Zitterbewegung* of electrons has been previously considered in graphene [58]. We expect the derived transient responses to be highly dependent on the disorder present in material, with the framework of this work being applicable only in the ultraclean system limit with extremely high scattering times τ . Nevertheless, we expect the derived intrinsic transient nonlinear

Zitterbewegung photocurrents to persist for any finite scattering times τ . The study of the effects of disorder and interactions on the quantum geometric nonlinear *Zitterbewegung* currents is left for future work.

Finally, we propose a dynamical protocol for extracting multistate torsion tensor and QGTs from transient current tomography. To reconstruct the multistate torsion, the Fourier components of the current at virtual band energy differences ω_{pn} , are of particular interest. Different drive frequencies ω and light polarizations allow to determine and control the amplitude factors $w_{nmp}^{+-}(\omega)$, $w_{nmp}^{\hat{x}\hat{x}}(\omega)$, beyond the filling factors f_m controllable with temperature and chemical potential. Acquisition of the dynamics of the Fourier modes with frequencies $\omega_{pn}(\mathbf{k})$ in the transient photocurrents thus allows to estimate T_{ijk}^{nmp} and Q_{ijj}^{nmp} averaged over a constant frequency (ω_{pn}) contour determined by the band energy splittings.

It should be noted that the zero-field (free) *Zitterbewegung* proportional to amplitudes Z_{ab} has been measured in ultracold atom quantum simulators [59]. More recently, a protocol to reconstruct the interband Berry connections that implicitly probes the *Zitterbewegung* amplitudes, $A_i^{nm} \propto Z_{nm}$, has been successfully developed in optical lattices [60]. In the latter, the shaking force F on optical lattice acts as an effective electric field that couples to the position operator, analogously to the Hamiltonian perturbation in the quantum Liouville equations. As a consequence, the nonlinear correction to the density matrix, $\rho_{nm}^{(2)} \propto F^2$, can be analogously induced, and its fingerprint in the nonlinear *Zitterbewegung* velocity $\xi_i^{m,(2)}(t)$ may be similarly measured.

To sum up, we elucidated the role of band torsion in the generalized equations of motion for the electron velocities in multiband systems. The multistate torsion and QGTs were shown to arise in the multimode *Zitterbewegung* induced by the second-order coupling to optical electric fields. The retrieved *Zitterbewegung* is manifested in nonlinear transient photocurrents, providing a direct off-resonant probe of multistate band torsion and QGTs.

Acknowledgements.— W.J.J. acknowledges funding from the Rod Smallwood Studentship at Trinity College, Cambridge. The author thanks G. Chaudhary, A. Jain, M. Mehraeen, G. Palumbo, and Anthropic's Claude, Fable 5.1, for helpful discussions.

-
- [1] P. Törmä, Essay: Where can quantum geometry lead us?, *Phys. Rev. Lett.* **131**, 240001 (2023).
 - [2] A. Bouhon, A. Timmel, and R.-J. Slager, *Quantum geometry beyond projective single bands* (2023), arXiv:2303.02180 [cond-mat.mes-hall].
 - [3] J. Yu, B. A. Bernevig, R. Queiroz, E. Rossi, P. Törmä, and B.-J. Yang, Quantum geometry in quantum materials, *npj Quantum Materials* **10**, 101 (2025).
 - [4] N. Verma, P. J. W. Moll, T. Holder, and R. Queiroz,

Quantum geometry and the hidden scales in materials, *Nat. Rev. Phys.* **8**, 226 (2026).

- [5] A. Gao, N. Nagaosa, N. Ni, and S.-Y. Xu, Quantum geometry phenomena in condensed matter systems, *Rev. Mod. Phys.* **98**, 035005 (2026).
- [6] S. Peotta and P. Törmä, Superfluidity in topologically nontrivial flat bands, *Nat. Commun.* **6**, 1 (2015).
- [7] F. Xie, Z. Song, B. Lian, and B. A. Bernevig, Topology-bounded superfluid weight in twisted bilayer graphene,

- Phys. Rev. Lett. **124**, 167002 (2020).
- [8] J. Herzog-Arbeitman, V. Peri, F. Schindler, S. D. Huber, and B. A. Bernevig, Superfluid weight bounds from symmetry and quantum geometry in flat bands, *Phys. Rev. Lett.* **128**, 087002 (2022).
 - [9] F. Piéchon, A. Raoux, J.-N. Fuchs, and G. Montambaux, Geometric orbital susceptibility: Quantum metric without Berry curvature, *Phys. Rev. B* **94**, 134423 (2016).
 - [10] C. W. Chau, R.-J. Slager, and W. J. Jankowski, Orbital magnetization reveals multiband topology, *Phys. Rev. B* **113**, 235154 (2026).
 - [11] R. Yuan, W. J. Jankowski, K. Shen, and R.-J. Slager, Quantum Geometry of Altermagnetic Magnons Probed by Light, *Phys. Rev. Lett.* **137**, 106901 (2026).
 - [12] H. Hu, O. Vafek, K. Haule, and B. A. Bernevig, Ferromagnetism versus antiferromagnetism in narrow-band systems: Competition between quantum geometry and band dispersion, *Phys. Rev. Lett.* **136**, 256505 (2026).
 - [13] W. Yao and Q. Niu, Berry Phase Effect on the Exciton Transport and on the Exciton Bose-Einstein Condensate, *Phys. Rev. Lett.* **101**, 106401 (2008).
 - [14] W. J. Jankowski, J. J. P. Thompson, B. Monserrat, and R.-J. Slager, Excitonic topology and quantum geometry in organic semiconductors, *Nat. Commun.* **16**, 4661 (2025).
 - [15] J. J. P. Thompson, W. J. Jankowski, R.-J. Slager, and B. Monserrat, Topologically enhanced exciton transport, *Nat. Commun.* **16**, 11448 (2025).
 - [16] H. Davenport, Y. Hwang, J. Knolle, and F. Schindler, Composite quantum geometry and semiclassical dynamics (2026), [arXiv:2606.12525 \[cond-mat.mes-hall\]](#).
 - [17] J.-H. Choi, S.-H. Han, Y.-K. Han, S.-W. Kim, J.-W. Rhim, and J. J. P. Thompson, Enhancement of exciton radius near a band-gap closing through quantum geometry (2026), [arXiv:2607.28731 \[cond-mat.mes-hall\]](#).
 - [18] N. Nagaosa, J. Sinova, S. Onoda, A. H. MacDonald, and N. P. Ong, Anomalous Hall effect, *Rev. Mod. Phys.* **82**, 1539 (2010).
 - [19] Y. Gao, S. A. Yang, and Q. Niu, Field Induced Positional Shift of Bloch Electrons and Its Dynamical Implications, *Phys. Rev. Lett.* **112**, 166601 (2014).
 - [20] I. Sodemann and L. Fu, Quantum nonlinear Hall effect induced by Berry curvature dipole in time-reversal invariant materials, *Phys. Rev. Lett.* **115**, 216806 (2015).
 - [21] J. Ahn, G.-Y. Guo, and N. Nagaosa, Low-frequency divergence and quantum geometry of the bulk photovoltaic effect in topological semimetals, *Phys. Rev. X* **10**, 041041 (2020).
 - [22] J. Ahn, G.-Y. Guo, N. Nagaosa, and A. Vishwanath, Riemannian geometry of resonant optical responses, *Nat. Phys.* **18**, 290–295 (2022).
 - [23] J. Mitscherling, A. Avdoshkin, and J. E. Moore, Gauge-invariant projector calculus for quantum state geometry and applications to observables in crystals, *Phys. Rev. B* **112**, 085104 (2025).
 - [24] Y. Onishi and L. Fu, Fundamental bound on topological gap, *Phys. Rev. X* **14**, 011052 (2024).
 - [25] S. Kwon and B.-J. Yang, Quantum geometric bound and ideal condition for Euler band topology, *Phys. Rev. B* **109**, L161111 (2024).
 - [26] W. J. Jankowski, A. S. Morris, Z. Davoyan, A. Bouhon, F. N. Ünal, and R.-J. Slager, Non-Abelian Hopf-Euler insulators, *Phys. Rev. B* **110**, 075135 (2024).
 - [27] W. J. Jankowski, A. S. Morris, A. Bouhon, F. N. Ünal, and R.-J. Slager, Optical manifestations and bounds of topological Euler class, *Phys. Rev. B* **111**, L081103 (2025).
 - [28] W. J. Jankowski, R.-J. Slager, and G. F. Lange, Quantum geometric bounds in spinful systems with trivial band topology, *Phys. Rev. Res.* **7**, L042011 (2025).
 - [29] W. J. Jankowski, R.-J. Slager, and M. Pizzochero, Enhancing the hyperpolarizability of crystals with quantum geometry, *Phys. Rev. Lett.* **135**, 126606 (2025).
 - [30] K. Shinada and N. Nagaosa, Quantum geometric bounds for observables: Linear responses, drude weight, and orbital magnetization, *Phys. Rev. B* **112**, 155158 (2025).
 - [31] Y.-C. Hung, Y. Onishi, H. Lin, L. Fu, and A. Bansil, Extending topological bound on quantum weight beyond symmetry-protected topological phases, *Phys. Rev. Lett.* **137**, 126601 (2026).
 - [32] J. P. Provost and G. Vallee, Riemannian structure on manifolds of quantum states, *Commun. Math. Phys.* **76**, 289 (1980).
 - [33] N. Marzari and D. Vanderbilt, Maximally localized generalized Wannier functions for composite energy bands, *Phys. Rev. B* **56**, 12847 (1997).
 - [34] R. Resta, The insulating state of matter: a geometrical theory, *The European Physical Journal B* **79**, 121 (2011).
 - [35] D. Xiao, M.-C. Chang, and Q. Niu, Berry phase effects on electronic properties, *Rev. Mod. Phys.* **82**, 1959 (2010).
 - [36] D. Vanderbilt, *Berry phases in electronic structure theory: electric polarization, orbital magnetization and topological insulators* (Cambridge University Press, 2018).
 - [37] M. Kang, S. Kim, Y. Qian, P. M. Neves, L. Ye, J. Jung, D. Puntel, F. Mazzola, S. Fang, C. Jozwiak, A. Bostwick, E. Rotenberg, J. Fuji, I. Vobornik, J.-H. Park, J. G. Checkelsky, B.-J. Yang, and R. Comin, Measurements of the quantum geometric tensor in solids, *Nat. Phys.* **21**, 110 (2025).
 - [38] S. Kim, Y. Chung, Y. Qian, S. Park, C. Jozwiak, E. Rotenberg, A. Bostwick, K. S. Kim, and B.-J. Yang, Direct measurement of the quantum metric tensor in solids, *Science* **388**, 1050 (2025).
 - [39] M. Mehraeen, Quantum response theory and momentum-space gravity, *Phys. Rev. Lett.* **135**, 156302 (2025).
 - [40] S. Farrag, E. Mele, and T. Low, High Order Geometric Channels for Nonlinear Transport in Bloch Bands (2026), [arXiv:2607.20702 \[cond-mat.mes-hall\]](#).
 - [41] W. J. Jankowski and R.-J. Slager, Quantized integrated shift effect in multigap topological phases, *Phys. Rev. Lett.* **133**, 186601 (2024).
 - [42] A. Avdoshkin, J. Mitscherling, and J. E. Moore, Multi-state geometry of shift current and polarization, *Phys. Rev. Lett.* **135**, 066901 (2025).
 - [43] Z. Guo, Z. Lu, H. Wang, and K. Chang, Bicircular light-induced multistate geometric current, *Phys. Rev. B* **112**, 035162 (2025).
 - [44] B. Bradlyn, Multi-state geometry of density matrices and rectification sum rules (2026), [arXiv:2608.06326 \[cond-mat.mes-hall\]](#).
 - [45] A. Jain, W. J. Jankowski, M. Mehraeen, and R.-J. Slager, Nonlinear Odd Viscoelastic Effect, *Phys. Rev. Lett.* **137**, 036302 (2026).
 - [46] A. Jain, W. J. Jankowski, M. Mehraeen, and R.-J. Slager, Topological acoustic diode (2026), [arXiv:2601.20951 \[cond-mat.mes-hall\]](#).
 - [47] Q. Tian, Z. Ni, M. Cothrine, D. G. Mandrus, E. J. Mele,

- A. M. Rappe, C. L. Kane, F. de Juan, and L. Wu, [Observation of magnetically switchable quantum geometric photocurrents](#) (2026), [arXiv:2605.22518 \[cond-mat.mtrl-sci\]](#).
- [48] A. Stern and F. von Oppen, [How quantum is quantum geometry?](#) (2026), [arXiv:2608.26269 \[cond-mat.mes-hall\]](#).
- [49] W. J. Jankowski, R.-J. Slager, and G. Palumbo, [Probing tensor monopoles and gerbe invariants in three-dimensional topological matter](#) (2025), [arXiv:2507.22116 \[cond-mat.mes-hall\]](#).
- [50] D. Culcer, Zitterbewegung velocity in semiclassical electron dynamics, [Phys. Rev. B](#) **114**, L020305 (2026).
- [51] See Supplemental Material (SM) for further details on the multistate quantum geometry (Sec. I), and the derivations of dynamically-induced Zitterbewegungs at first order (Sec. II) and second order (Sec. III) in optical electric fields.
- [52] N. Verma and R. Queiroz, Instantaneous response and quantum geometry of insulators, [Proceedings of the National Academy of Sciences](#) **122**, e2405837122 (2025).
- [53] D. Mandal, S. Sarkar, K. Das, and A. Agarwal, Quantum geometry induced third-order nonlinear transport responses, [Phys. Rev. B](#) **110**, 195131 (2024).
- [54] T. Bzdušek, From quantum geometry to nonlinear optics and gerbes: Recent advances in topological band theory, [Phys. Rev. B](#) **113**, 099601 (2026).
- [55] G. Dávid and J. Cserti, General theory of Zitterbewegung, [Phys. Rev. B](#) **81**, 121417(R) (2010).
- [56] Cserti, József and Dávid, Gyula, Relation between Zitterbewegung and the charge conductivity, Berry curvature, and the Chern number of multiband systems, [Phys. Rev. B](#) **82**, 201405(R) (2010).
- [57] N. Verma and R. Queiroz, Framework to measure quantum metric from step response, [Phys. Rev. Lett.](#) **134**, 106403 (2025).
- [58] T. M. Rusin and W. Zawadzki, Multimode behavior of electron Zitterbewegung induced by an electromagnetic wave in graphene, [Phys. Rev. B](#) **88**, 235404 (2013).
- [59] J. Y. Vaishnav and C. W. Clark, Observing Zitterbewegung with Ultracold Atoms, [Phys. Rev. Lett.](#) **100**, 153002 (2008).
- [60] S.-W. Chang, M. N. Schwarz, E. G. Moloney, K. Lin, and D. M. Stamper-Kurn, [Interband Berry connection measurement in the optical honeycomb lattice](#) (2026), [arXiv:2605.11597 \[cond-mat.quant-gas\]](#).

SUPPLEMENTAL MATERIAL
Multistate Torsion in *Zitterbewegung*

Wojciech J. Jankowski^{1, *}

¹*Theory of Condensed Matter Group, Cavendish Laboratory, University of Cambridge, J. J. Thomson Avenue, Cambridge CB3 0HE, UK*
(Dated: September 30, 2026)

CONTENTS

I. Details on the multistate quantum geometry	2
II. Derivation of the first-order field-induced <i>Zitterbewegung</i> velocity	2
III. Derivation of the second-order field-induced <i>Zitterbewegung</i> velocity	4
References	6

* wjj25@cam.ac.uk

I. DETAILS ON THE MULTISTATE QUANTUM GEOMETRY

We begin with a set of definitions. $f_{mn} \equiv f_m - f_n$ denotes band occupation factor differences, $\omega_{mn} = (\varepsilon_m - \varepsilon_n)/\hbar$ are the frequencies given by the band energy differences. In this work, we consider nondegenerate bands, i.e., $\varepsilon_m \neq \varepsilon_n$, for $m \neq n$. We introduce momentum derivatives, $\partial_i \equiv \partial_{k_i}$, differences of intraband group velocities $\Delta_i^{nm} = \partial_i \omega_{nm}$, and non-Abelian Berry connections, $A_i^{nm} = i \langle u_{n\mathbf{k}} | \partial_i u_{m\mathbf{k}} \rangle$, of Bloch states, $|\psi_{n\mathbf{k}}\rangle = e^{i\mathbf{k}\cdot\mathbf{r}} |u_{n\mathbf{k}}\rangle$, where $\mathbf{r}$ is the position operator with components $\hat{r}_i$. The interband velocities v_i^{nm} ($n \neq m$) are given by the non-Abelian Berry-connection elements,

$$v_i^{nm} = \langle u_{n\mathbf{k}} | \hat{v}_i | u_{m\mathbf{k}} \rangle = \langle u_{n\mathbf{k}} | \frac{i}{\hbar} [H_{\mathbf{k}}, \hat{r}_i] | u_{m\mathbf{k}} \rangle = i\omega_{nm} A_i^{nm}, \quad (1)$$

as the position operator matrix elements read: $\langle \psi_{n\mathbf{k}} | \hat{r}_i | \psi_{m\mathbf{k}'} \rangle = -i\delta_{nm} \partial_i \delta(\mathbf{k} - \mathbf{k}') + A_i^{nm}$ [1]. We define covariant derivatives as,

$$\mathcal{D}_i^{mn} \equiv \partial_i - i(A_i^{mm} - A_i^{nn}). \quad (2)$$

The two-state quantum geometric tensors (QGTs) read [1],

$$\mathcal{Q}_{ij}^{nm} = A_i^{nm} A_j^{mn} = g_{ij}^{nm} - \frac{i}{2} F_{ij}^{nm}, \quad (3)$$

with a two-state quantum metric g_{ij}^{nm} and two-state symplectic form F_{ij}^{nm} representing a band-projected contribution to a Berry curvature [1]. The Hermitian connections can be defined as [1],

$$C_{ijk}^{nm} = A_i^{nm} \mathcal{D}_i^{mn} A_j^{nn}, \quad (4)$$

which culminates in a two-state torsion tensor T_{ijk}^{nm} [1, 2],

$$T_{ijk}^{nm} = C_{ijk}^{nm} - C_{ikj}^{nm} = iA_i^{nm} \sum_{p \neq n, m} (A_j^{mp} A_k^{pn} - A_k^{mp} A_j^{pn}) = \sum_{p \neq n, m} T_{ijk}^{nmp}. \quad (5)$$

T_{ijk}^{nmp} is a multistate torsion tensor component central to this work,

$$T_{ijk}^{nmp} = iA_i^{nm} (A_j^{mp} A_k^{pn} - A_k^{mp} A_j^{pn}) = i(\mathcal{Q}_{ijk}^{nmp} - \mathcal{Q}_{ikj}^{nmp}). \quad (6)$$

In the above, we further defined a three-state QGT [3],

$$\mathcal{Q}_{ijk}^{nmp} = A_i^{nm} A_j^{mp} A_k^{pn}. \quad (7)$$

It should be noted that the three-state torsion tensor component T_{ijk}^{nmp} and the three-state QGTs, $\mathcal{Q}_{ijk}^{nmp}$, are gauge-invariant under the $U(1)$ gauge transformations of the individual nondegenerate Bloch bands. Hence, these tensors constitute measurable multistate quantum geometric quantities, as demonstrated in the main text.

II. DERIVATION OF THE FIRST-ORDER FIELD-INDUCED ZITTERBEWEGUNG VELOCITY

To derive the field-induced first-order *Zitterbewegung* velocity, we begin with the quantum Liouville equation for a single-particle density matrix $\rho(t)$,

$$\partial_t \rho(t) + \frac{i}{\hbar} [H(t), \rho(t)] = 0. \quad (8)$$

The time-dependent Hamiltonian, $H(t) = H_{\mathbf{k}} + \Delta H(t)$, is given by an unperturbed free-fermion Bloch Hamiltonian $H_{\mathbf{k}}$ under a minimal coupling to an electric field perturbation in length gauge, $\Delta H(t) = e\mathbf{r} \cdot \mathbf{E}(t)$. $\mathbf{r}$ denotes a single-particle position operator, $\mathbf{E}(t)$ represents an electric field vector. We expand the density matrix perturbatively to N -th order in the electric field powers as: $\rho(t) = \rho^{(0)}(t) + \rho^{(1)}(t) + \rho^{(2)}(t) + \dots + \rho^{(N)}(t)$, which yields a hierarchy of equations [4],

$$\partial_t \rho^{(N)}(t) + \frac{i}{\hbar} [H_{\mathbf{k}}, \rho^{(N)}(t)] + \frac{\rho^{(N)}(t)}{\tau^{(N)}} = \frac{i\mathbf{E}(t)}{\hbar} \cdot [\mathbf{r}, \rho^{(N-1)}]. \quad (9)$$

$\tau^{(N)}$ are scattering times which we send to $\tau^{(N)} \rightarrow \infty$ in the clean crystal limit central to this work. In the band basis, $\rho_{nm}^{(0)}(t) = \langle u_{n\mathbf{k}} | \rho^{(0)}(t) | u_{m\mathbf{k}} \rangle$, with Bloch states, $|\psi_{n\mathbf{k}}\rangle = e^{i\mathbf{k}\cdot\mathbf{r}} |u_{n\mathbf{k}}\rangle$, satisfying the unperturbed Schrödinger equation, $H_{\mathbf{k}} |u_{n\mathbf{k}}\rangle = \varepsilon_n |u_{n\mathbf{k}}\rangle$.

The zero-field density matrix, $\rho_{nm}^{(0)}(t) = \delta_{nm}f_m$, is given by the Fermi–Dirac distribution f_m . We focus on the field-induced first-order and second-order corrections: $\rho_{nm}^{(1)}(t) \equiv \rho_E^{nm}(t)$ and $\rho_{nm}^{(2)}(t) \equiv \rho_{E^2}^{nm}(t)$ in what follows.

We choose an instantaneous electric field of monochromatic light, $E_j(t) = \Theta(t)\text{Re}[E_j(\omega)e^{i\omega t}]$, where $\Theta(t)$ is a Heaviside step function. Accordingly, $\rho_{nm}^{(1)}(0) = 0$. $\rho_{nm}^{(1)}(t)$ for $t > 0$ follows from the quantum Liouville equation as,

$$\partial_t \rho_E^{mn}(t) + i\omega_{nm}\rho_E^{nm}(t) = \frac{ie}{\hbar} A_j^{nm} f_{mn} E_j(t). \quad (10)$$

The solution to the Liouville equation reads,

$$\rho_E^{mn}(t) = \frac{e}{2} A_j^{nm} f_{mn} \left[E_j \frac{e^{-i\omega t} - e^{-i\omega_{mn}t}}{\varepsilon_m - \varepsilon_n - \hbar\omega} + E_j^* \frac{e^{i\omega t} - e^{-i\omega_{mn}t}}{\varepsilon_m - \varepsilon_n + \hbar\omega} \right]. \quad (11)$$

Combining the time-dependent field-induced coherence $\rho_E^{mn}(t)$ with the velocity operator as $v_i^{nm} \rho_E^{mn}(t) \rightarrow \xi_i^{(1),m}(t)f_m$, and isolating the oscillatory time-dependent term, gives the first-order field-induced *Zitterbewegung* velocity [5],

$$\xi_i^{(1),m}(t) = \frac{ieE_j(\omega)}{\hbar} \sum_{n \neq m} Q_{ij}^{mn} \frac{\omega_{mn}}{\omega_{mn} - \omega} e^{-i\omega_{mn}t} + \text{h.c.} \quad (12)$$

We note that the dc limit $\omega \rightarrow 0$ restores the static field result of first-order field-induced *Zitterbewegung* velocity of Ref. [5].

Furthermore, it is interesting to consider the field-induced first-order *Zitterbewegung* under a circularly polarized drive. The corresponding monochromatic field with amplitude $|E_0|$ reads: $\mathbf{E}(t) = \Theta(t) \frac{E_0}{2} \sum_{\eta=\pm} \boldsymbol{\epsilon}^{(\eta)} e^{i\eta\omega t}$ with $\boldsymbol{\epsilon}^{(\pm)} = \frac{1}{\sqrt{2}}(1, \pm i, 0)$ representing the left- and right- circular light polarization vectors. We consider light polarized circularly in the $x-y$ plane. For convenience, we define,

$$A_{\pm}^{mn} = A_x^{mn} + iA_y^{mn}. \quad (13)$$

Correspondingly, solving the Liouville equation to first order in electric field yields,

$$\rho_E^{mn}(t) = \frac{eE_0}{2} f_{mn} \left[A_+^{mn} \frac{e^{-i\omega t} - e^{-i\omega_{mn}t}}{\varepsilon_m - \varepsilon_n - \hbar\omega} + A_-^{mn} \frac{e^{i\omega t} - e^{-i\omega_{mn}t}}{\varepsilon_m - \varepsilon_n + \hbar\omega} \right], \quad (14)$$

for left circularly polarized electric field, $\mathbf{E}(t) = \Theta(t)E_0[\hat{x} \cos(\omega t) + \hat{y} \sin(\omega t)]$, with $\mathbf{E}(\omega) = \frac{E_0}{2}(1, i, 0)$. The resultant circular field-induced *Zitterbewegung* amounts to,

$$\xi_i^{(1),m}(t) = -\frac{eE_0}{\hbar} \sum_{n \neq m} \omega_{mn} \text{Im} \left[\left(\frac{Q_{i+}^{mn}}{\omega_{mn} - \omega} + \frac{Q_{i-}^{mn}}{\omega_{mn} + \omega} e^{-i\omega_{mn}t} \right) \right], \quad (15)$$

where we defined circularly-projected two-state QGTs, $Q_{i\pm}^{mn} = A_{\pm}^{mn} A_{\pm}^{nm}$. Finally, it is interesting to integrate the first-order field-induced *Zitterbewegung* velocity, $\Delta r_i^m(t) \equiv \int_0^t dt' \xi_i^{(1),m}(t')$, in the $\omega \neq 0$ regime, to compare with the field-induced positional shift [6] arising from the dc electric field limit of the field-induced *Zitterbewegung* velocity. For the circularly polarized drive, the integration of the electric field amounts to an impulse $\mathcal{I}(t) \equiv \int_0^t dt' \mathbf{E}(t')$, and we obtain,

$$\Delta r_i^m(t) = 2\text{Re} [\alpha_{ij}^m(\omega) E_j(\omega) e^{-i\omega t}] - \frac{e\mathcal{I}_j(t)}{\hbar} \sum_{n \neq m} f_{mn} \Omega_{ij}^{mn} - 2\text{Re} \left[\frac{e}{\hbar} \sum_{n \neq m} f_{mn} Q_{ij}^{mn} \left(\frac{E_j(\omega)}{\omega_{mn} - \omega} + \frac{E_j^*(\omega)}{\omega_{mn} + \omega} \right) e^{-i\omega_{mn}t} \right]. \quad (16)$$

The first term is an ac generalization of the field-induced positional shift of a Bloch state [6],

$$\alpha_{ij}^m(\omega) = \frac{e}{\hbar} \sum_{n \neq m} f_{mn} \left[\frac{Q_{ij}^{mn}}{\omega_{mn} - \omega} + \frac{Q_{ij}^{nm}}{\omega_{mn} + \omega} \right] = \frac{e}{\hbar} \sum_{n \neq m} f_{mn} \frac{2\omega_{mn} g_{ij}^{nm} - i\omega \Omega_{ij}^{mn}}{\omega_{mn}^2 - \omega^2}, \quad (17)$$

which corresponds to the static field-induced position shift, $\alpha_{ij}^m(0) = \frac{2e}{\hbar} \frac{g_{ij}^{nm}}{\omega_{mn}}$, in the dc limit, $\omega \rightarrow 0$ [5, 6].

In the next section, we derive the second-order field-induced *Zitterbewegung* velocity, upon extending the above developments derived from the quantum Liouville equation.

III. DERIVATION OF THE SECOND-ORDER FIELD-INDUCED ZITTERBEWEGUNG VELOCITY

Below, we derive the second-order field-induced *Zitterbewegung* velocity. The first-order field-induced interband coherences in the density matrix, $\rho_{nm}^{(1)}(t) \equiv \rho_E^{nm}(t)$, under circularly polarized monochromatic drive were considered in the previous section. We now focus on the second-order field-induced interband coherences in the density matrix, $\rho_{nm}^{(2)}(t) \equiv \rho_{E^2}^{nm}$, induced by the circularly polarized monochromatic drive with frequency ω .

We again choose a monochromatic field with amplitude $|E_0|$ and electric field vector, $\mathbf{E}(t) = \Theta(t) \frac{E_0}{2} \sum_{\eta=\pm} \boldsymbol{\epsilon}^{(\eta)} e^{i\eta\omega t}$, with $\boldsymbol{\epsilon}^{(\pm)} = \frac{1}{\sqrt{2}}(1, \pm i, 0)$ representing the left- and right- circular light polarization vectors. The quantum Liouville equation at the second perturbative order reads,

$$\partial_t \rho_{E^2}^{mn}(t) + i\omega_{nm} \rho_{E^2}^{mn}(t) = \frac{eE_i(t)}{\hbar} [\mathcal{D}_i \rho_E(t)]^{mn}, \quad (18)$$

with the covariant derivative acting on the density matrix as [4],

$$[\mathcal{D}_i \rho]^{mn} = \partial_i \rho^{mn} - i[A_i, \rho]^{mn}. \quad (19)$$

$A_i = \sum_{n,m} A_i^{nm} |u_{nk}\rangle \langle u_{mk}|$ is the Berry connection operator. Correspondingly, we employ the results for the first-order density matrix interband coherences $\rho_E^{nm}(t)$ under the circular drive, as derived in the previous section. The first-order density matrix corrections in the clean limit ($\tau \rightarrow \infty$) read,

$$\rho_E^{mn}(t) = \frac{eE_0}{2\hbar} \sum_{\eta} i(\boldsymbol{\epsilon}^{(\eta)} \cdot \partial_{\mathbf{k}}) f_{nm} \frac{1 - e^{-i\eta\omega t}}{i\eta\omega}, \quad (20)$$

$$\rho_E^{nm}(t) = \frac{eE_0}{2\hbar} \sum_{\eta} i(\boldsymbol{\epsilon}^{(\eta)} \cdot \mathbf{A}^{nm}) f_{nm} \frac{e^{-i\eta\omega t} - e^{-i\omega_{nm}t}}{i(\omega_{nm} - \eta\omega)}, \quad (21)$$

with the Berry connection vector $\mathbf{A}^{nm} = (A_x^{nm}, A_y^{nm}, A_z^{nm})$. Under the circular polarizations η, η' , we combine electric fields as, $E_c E_d \sim \frac{E_0^2}{4} \boldsymbol{\epsilon}_c^{(\eta)} \boldsymbol{\epsilon}_d^{(\eta')}$. We first obtain $\rho_{E^2}^{mn}(t)$ upon solving the Liouville equation in the dc limit ($\omega \rightarrow 0$) as,

$$\begin{aligned} \rho_{E^2}^{mn}(t) = \frac{e^2 E_c E_d}{\hbar^2} & \left[i(\mathcal{D}_c^{nm} A_d^{mn}) f_{mn} \left(-\frac{1 - e^{-i\omega_{nm}t}}{\omega_{nn}^2} + \frac{ite^{-i\omega_{nm}t}}{\omega_{mn}} \right) + iA_d^{mn} f_{mn} \Delta_c^{mn} \left(\frac{t^2 e^{-i\omega_{nm}t}}{2\omega_{mn}} - \frac{ite^{-i\omega_{nm}t}}{\omega_{mn}^2} + \frac{1 - e^{-i\omega_{nm}t}}{\omega_{mn}^3} \right) \right. \\ & \left. + iA_d^{mn} \partial_c f_{mn} \left(-\frac{1 - e^{-i\omega_{nm}t}}{\omega_{mn}^2} + \frac{ite^{-i\omega_{nm}t}}{\omega_{mn}} \right) + iA_c^{mn} \partial_d f_{mn} \left(-\frac{it}{\omega_{mn}} + \frac{1 - e^{-i\omega_{nm}t}}{\omega_{mn}^2} \right) + \rho_{E^2,3b}^{mn}(t) \right], \end{aligned} \quad (22)$$

with the multistate three-band term $\rho_{E^2,3b}^{mn}(t)$,

$$\rho_{E^2,3b}^{mn}(t) = \frac{e^2 E_c E_d}{\hbar^2} \sum_{p \neq n, m} \left[A_c^{mp} A_d^{pn} f_{pn} L_{mn,pn}(t) - A_d^{mp} A_c^{pn} f_{mp} L_{nm,np}(t) \right]. \quad (23)$$

The three-state coherence term includes a time-dependent kernel $L_{mn,pn}(t)$,

$$L_{mn,pn}(t) = -\frac{1 - e^{-i\omega_{mn}t}}{\omega_{pn}\omega_{mn}} + \frac{e^{-i\omega_{pn}t} - e^{-i\omega_{mn}t}}{\omega_{pn}\omega_{mp}}. \quad (24)$$

We note that this multiband coherence term is consistent with the multistate $\rho_{nm}^{(2)}$ derived in Ref. [4]. Accordingly, the multistate contribution to the *Zitterbewegung* velocity in the static field limit reads,

$$\xi_{3b,i}^{(2),m}(t) = \frac{e^2 E_c E_d}{\hbar^2} \sum_{n \neq p \neq m} \left[-i\omega_{ln} Q_{icd}^{npm} L_{pn,mn}(t) + i\omega_{nm} Q_{icd}^{nmp} L_{nm,pm}(t) - i\omega_{nm} Q_{idc}^{npm} L_{pn,pm}(t) + i\omega_{mn} Q_{idc}^{nmp} L_{mn,mp}(t) \right]. \quad (25)$$

We now extend these results to finite frequencies ($\omega \neq 0$) for photons with circular polarizations η, η' . The dynamically-induced second-order *Zitterbewegung* velocity is derived analogously upon solving the quantum Liouville equations. To state the result compactly, we define a set of frequency-based factors,

$$L_{mn,pn}^{\eta'\eta}(t) = -\frac{1}{\omega_{pn} - \eta\omega} \left[\frac{e^{-i(\eta+\eta')\omega t} - e^{-i\omega_{mn}t}}{\omega_{mn} - (\eta + \eta')\omega} - \frac{e^{-i(\omega_{pn}+\eta'\omega)t} - e^{-i\omega_{mn}t}}{\omega_{mn} - \omega_{pn} - \eta'\omega} \right], \quad (26)$$

$$K_{mn}^{\eta'\eta}(t) = -\frac{1}{\omega_{mn} - \eta\omega} \left[\frac{e^{-i(\eta+\eta')\omega t} - e^{-i\omega_{mn}t}}{\omega_{mn} - (\eta + \eta')\omega} - e^{-i\omega_{mn}t} \frac{1 - e^{-i\eta'\omega t}}{\eta'\omega} \right], \quad (27)$$

$$J_{mn}^{\eta'\eta}(t) = \frac{e^{-i\omega_{mn}t}}{\omega_{mn} - \eta\omega} \left[\frac{it e^{-i\eta'\omega t}}{\eta'\omega} - \frac{1 - e^{-i\eta'\omega t}}{\omega^2} \right] - \frac{K_{mn}^{\eta'\eta}}{\omega_{mn} - \eta\omega}, \quad (28)$$

$$M_{mn}^{\eta'\eta}(t) = \frac{1}{\omega_{mn} - \eta\omega} \left[\frac{e^{-i(\eta+\eta')\omega t} - 1}{(\eta + \eta')\omega} - \frac{e^{-i(\omega_{mn}+\eta'\omega)t} - 1}{\omega_{mn} + \eta'\omega} \right], \quad (29)$$

$$N_{mn}^{\eta'\eta}(t) = \frac{1}{i\eta\omega} \left[\frac{e^{-i\eta'\omega t} - e^{-i\omega_{mn}t}}{i(\omega_{mn} - \eta'\omega)} - \frac{e^{-i(\eta+\eta')\omega t} - e^{-i\omega_{mn}t}}{i(\omega_{mn} - (\eta + \eta')\omega)} \right]. \quad (30)$$

Correspondingly, the dynamically-induced second-order *Zitterbewegung* velocity is given as,

$$\begin{aligned} \xi_i^{(2),m}(t) = & \frac{e^2 E_c E_d}{4\hbar^2} \sum_{\eta,\eta'} \sum_{n \neq m} \left[\omega_{mn} C_{icd}^{mn} K_{mn}^{\eta'\eta}(t) + \omega_{mn} Q_{id}^{mn} \Delta_{mn}^c J_{mn}^{\eta'\eta}(t) - (m \leftrightarrow n) \right] + \sum_{n \neq p} v_m^i \left[Q_{cd}^{mn} M_{mn}^{\eta'\eta}(t) - (m \leftrightarrow n) \right] \\ & + \sum_{n \neq p \neq m} \left[-i\omega_{pn} Q_{icd}^{npm} L_{pn,mn}^{\eta'\eta}(t) + i\omega_{nm} Q_{icd}^{mnp} L_{nm,pn}^{\eta'\eta}(t) - i\omega_{pn} Q_{idc}^{npm} L_{pn,pn}^{\eta'\eta}(t) + i\omega_{mn} Q_{idc}^{mnp} L_{mn,mp}^{\eta'\eta}(t) \right] + \dots, \end{aligned} \quad (31)$$

where ... includes additional terms derived upon integrating by parts distribution function nonuniformities, $\partial_k f_{mn}$. The finite-frequency multistate *Zitterbewegung* term reads,

$$\xi_{3b,i}^{(2),m}(t) = \frac{e^2 E_c E_d}{\hbar^2} \sum_{n \neq p \neq m} \left[-i\omega_{pn} Q_{icd}^{npm} L_{pn,mn}^{\eta'\eta}(t) + i\omega_{nm} Q_{icd}^{mnp} L_{nm,pn}^{\eta'\eta}(t) - i\omega_{pn} Q_{idc}^{npm} L_{pn,pn}^{\eta'\eta}(t) + i\omega_{mn} Q_{idc}^{mnp} L_{mn,mp}^{\eta'\eta}(t) \right]. \quad (32)$$

We will now focus on the contribution at the frequency ω_{pn} corresponding to the energy difference of virtual levels,

$$\xi_i^{(2),m}(t) \Big|_{\omega_{pn}} \propto \frac{e^2 E_0^2}{\hbar^2} \mathcal{A}_i^{mnp}(\omega) e^{-i\omega_{pn}t}. \quad (33)$$

The frequency-dependent amplitude factor consistent with Eq. (32) reads,

$$\mathcal{A}_i^{mnp}(\omega) = \frac{i}{4} A_i^{np} \sum_{\eta,\eta'} \frac{\omega_{pn}}{\omega_{pn} - (\eta + \eta')\omega} \left[\frac{(\epsilon^{(\eta')}) \cdot \mathbf{A}^{pm} (\epsilon^{(\eta)}) \cdot \mathbf{A}^{mn}}{\omega_{pm} - \eta'\omega} + \frac{(\epsilon^{(\eta)}) \cdot \mathbf{A}^{pm} (\epsilon^{(\eta')}) \cdot \mathbf{A}^{mn}}{\omega_{mn} - \eta'\omega} \right]. \quad (34)$$

Correspondingly, we can identify *Zitterbewegung* velocities $\xi_i^{(2),m}(t) \Big|_{\omega_{pn}}^{\eta\eta'}$ arising in the individual helicity channels ($\eta, \eta' = \pm$),

$$\xi_i^{(2),m}(t) \Big|_{\omega_{pn}}^{+-} = \frac{e^2 E_0^2}{\hbar^2} w_{mnp}^{+-}(\omega) \left[\text{Im } T_{ixy}^{nmp} \cos(\omega_{pn}t) + \text{Re } T_{ixy}^{nmp} \sin(\omega_{pn}t) \right], \quad (35)$$

for the counter-rotating amplitude, $\xi_i^{(2),m}(t) \propto \mathcal{A}_{ijk}^{mnp}(\omega, -\omega) E_j(\omega) E_k(-\omega)$, using the main text notation. The frequency-dependent factor reads,

$$w_{mnp}^{+-}(\omega) = \frac{\omega}{\omega_{mn}^2 - \omega^2} - \frac{\omega}{\omega_{pm}^2 - \omega^2}. \quad (36)$$

The co-rotating helicity channel yields an amplitude $\mathcal{A}_{ijk}^{mnp}(\omega, \omega)$, such that $\xi_i^{(2),m}(t) \propto \mathcal{A}_{ijk}^{mnp}(\omega, \omega) E_j(\omega) E_k(\omega)$, which culminates in a *Zitterbewegung* velocity given by,

$$\xi_i^{(2),m}(t) \Big|_{\omega_{pn}}^{++} = \frac{e^2 E_0^2}{\hbar^2} \text{Re} \left[w_{mnp}^{++}(\omega) (Q_{ixy}^{npm} + Q_{ixy}^{nmp}) e^{-i\omega_{pn}t} \right], \quad (37)$$

with another frequency-dependent factor,

$$w_{mnp}^{++}(\omega) = \frac{1}{\omega_{pn} - 2\omega} \left(\frac{1}{\omega_{pm} - \omega} - \frac{1}{\omega_{mn} - \omega} \right). \quad (38)$$

Interestingly, on choosing a linear polarization of photons instead, $\epsilon = \frac{1}{\sqrt{2}}(\epsilon^{(+)} + \epsilon^{(-)}) = \hat{x}$, the dynamically-induced nonlinear *Zitterbewegung* velocity component arises from three-state QGTs Q_{ixx}^{nmp} as,

$$\xi_i^{(2),m}(t) \Big|_{\omega_{pn}}^{\hat{x}\hat{x}} = -\frac{e^2 E_0^2}{\hbar^2} \sum_{n,p} w_{mnp}^{\hat{x}\hat{x}}(\omega) \left[\text{Im } Q_{ixx}^{nmp} \cos(\omega_{pn}t) - \text{Re } Q_{ixx}^{nmp} \sin(\omega_{pn}t) \right]. \quad (39)$$

The multiband frequency factor $w_{mnp}^{\hat{x}\hat{x}}(\omega)$ reads,

$$w_{mnp}^{\hat{x}\hat{x}}(\omega) = \sum_{\eta=\pm} \frac{\omega_{pn} - \eta\omega}{(\omega_{pm} - \eta\omega)(\omega_{mn} - \eta\omega)}. \quad (40)$$

Therefore, the nonlinear *Zitterbewegung* velocities that arise at second order in electric field coupling provide a window to measure both the multistate torsion tensors T_{ijk}^{nmp} and the three-state QGTs Q_{ijj}^{nmp} under the optical drives with distinct light polarizations.

-
- [1] J. Ahn, G.-Y. Guo, N. Nagaosa, and A. Vishwanath, *Nat. Phys.* **18**, 290–295 (2022).
 - [2] W. J. Jankowski and R.-J. Slager, *Phys. Rev. Lett.* **133**, 186601 (2024).
 - [3] M. Mehraeen, *Phys. Rev. Lett.* **135**, 156302 (2025).
 - [4] D. Mandal, S. Sarkar, K. Das, and A. Agarwal, *Phys. Rev. B* **110**, 195131 (2024).
 - [5] D. Culcer, *Phys. Rev. B* **114**, L020305 (2026).
 - [6] Y. Gao, S. A. Yang, and Q. Niu, *Phys. Rev. Lett.* **112**, 166601 (2014).